

Problem Set 4

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1 Question 1

- a The first regression implies that age is a determinant in explaining wage, while the second one would tell us the opposite. The relationship between how wage could determine age. I would say the first is more consistent with economic theory since we keep in mind that the relationship between wage and age is not that a higher wage implies that we magically become older but instead that as we get older our wage increases due to experience in the labor market.

- b i The first one is true since as we can see $\frac{1}{.1172468} \neq .0110603$. This is mathematically done by,

$$\frac{wage_i - \alpha_i - \epsilon_i}{\beta_1} = age_i$$

However, if we look at what age is equal to

$$age_i = \alpha_2 + \beta_2 wage_i + \epsilon_i$$

Thus, for this to hold we would need the intercepts equal to zero as well as the error term. Which is an unreasonable assumption.

- ii Since RSS is dependent on variance of its predictor then we can notice the following two formulas.

$$RSS_{wage} = (1 - r^2)var(wage)$$

$$RSS_{age} = (1 - r^2)var(age)$$

Thus since both of these depend on different variances then they are clearly not going to be equal unless, the variances are equal. The rest, follow this idea for TSS and ESS.

For ESS

$$\hat{y}_{wage} \neq \hat{y}_{age}$$

thus when calculating it on the average input for it's own regression it will not equal each other. Especially since they are based on different errors thus they cant be equal.

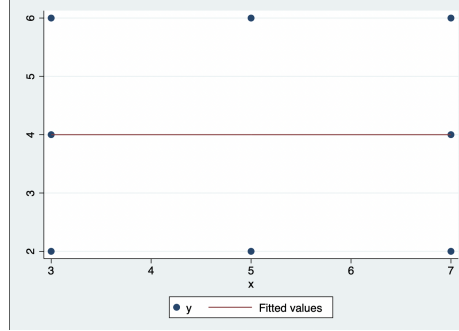


Figure 1: Fitted Reg for Data Set 1

- iii Since R^2 is calculated as the proportion of variance explained, and the numerator involves the square of the covariance $\sum_i (x_i - \bar{x})(y_i - \bar{y})$, swapping x and y does not change this value. Mathematically, we have

$$R_{wage}^2 = \frac{\hat{\beta}_1^2 \sum_i (age_i - \bar{age})^2}{\sum_i (wage_i - \bar{wage})^2} = \frac{[\sum_i (age_i - \bar{age})(wage_i - \bar{wage})]^2}{\sum_i (age_i - \bar{age})^2 \sum_i (wage_i - \bar{wage})^2} = R_{age}^2$$

Thus, the R^2 values are equal for both regressions.

- iv The t -statistics for the slope coefficients are calculated by

$$t_{\hat{\beta}_1} = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}, \quad t_{\hat{\beta}_2} = \frac{\hat{\beta}_2}{SE(\hat{\beta}_2)}$$

Since in simple regression the t -statistic is related to R^2 by

$$t^2 = F = \frac{R^2}{1 - R^2} \cdot (n - 2),$$

and we already showed that $R_{wage}^2 = R_{age}^2$, it follows that

$$t_{\hat{\beta}_1} = t_{\hat{\beta}_2}.$$

Thus, even though the slopes are different, the t -statistics are equal.

- c That a big part of finding causal relationship is really understanding that x is the one that affects y and not the other way around. That the issue might be of reverse causality because we wouldn't say that your wage affects your age but instead that age is the true determinant of wage.

2 Question 2

We see that the R^2 for this data set is 0. As we can see in figure 1 we end up seeing that the fitted values vs our y and x where we have the same amount of points above our regression line as below.

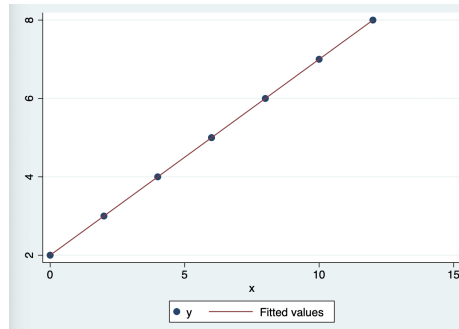


Figure 2: Fitted Reg for Data Set 2

3 Question 3

We see that the R^2 for this data set is 1. As we can see in figure 2 it is because our x and y follow a linear relation.

4 Question 4

Yes, mathematically, if we just add an infinite amount of statistically important variables, we would still find a higher R^2 . However, from the example from class we could also have an issue that some of the variables together explain a big part of our variable of interest, however, if they also explain the same parts then our regression would fit them as insignificant even though they help explain the relationship.

5 Question 5

- The first intercept α_1 is the average for males, while β_1 ends up being the difference between the average for males vs females. Now for the second intercept α_2 is the average for female where β_2 is the difference in the averages with female as the base group and male.
- The relationship ends up being that they get the same magnitude but opposite signs.

$$\beta_1 = \bar{w}_f - \bar{w}_m$$

$$\beta_2 = \bar{w}_m - \bar{w}_f$$

Thus we can see that it ends up being the same subtraction however flipped.

- Since we know from part b that

$$\beta_1 = \bar{w}_f - \bar{w}_m$$

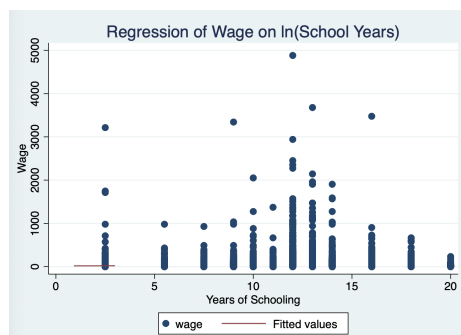


Figure 3: Graph

thus we could say

$$\beta_1 + \bar{w}_m = \bar{w}_f$$

and since we know that the intercepts are just the averages for male and female we could say.

$$\beta_1 + \alpha_1 = \alpha_2$$

- d Since they have the same standard errors and we noticed the relationship from before we end up just with opposite signs for the t statistics.
- e They are zero and this is because since we are just regressing our intercepts on a dummy variable and we are getting averages. Then half of the points will be above and half of them will be below thus giving us a R^2 of zero.

6 Question 6

- a This one it can't estimate $female^2$ because it is perfectly colinear with female since we are just squaring ones and zeros
- b It can't estimate the $\ln(female)$ because since we have zeros the log of zeros will give us errors when taking the regression
- c We can't calculate $\ln age$ and $\ln age^2$ since they are both perfectly correlated, this is due to us having one be determined by the other and since we logged age 2 it doesn't bring anymore quadratic information.
- d This is the same problem as before since our experience variable was estimated using age and scholar thus having perfect multicollinearity and stat dropping it.

7 Question 7

- a

b The intercept is for a person with one year of schooling.

c i Using the code

```
. display 20 - (_b[_cons] + _b[lnschoolyr]*log(17))  
-.58563003
```

```
ii . display _b[lnschoolyr] / 17  
.08277422
```

```
iii . display 0.01 * _b[lnschoolyr] * 100  
1.4071617
```

8 Question 8

a Region is formatted wrong since we are going to have a problem, the best example is, if we put zero in for our region variable who are we taking about? It would lead to a person that doesn't live in our world. Also it would lead to incorrect and biased inferences.

b I would write the model

$$wage_i = \beta_1^1 + \beta_2^1 female_i + \beta_3^1 schoolyr_i + \beta_4^1 exp_i + \beta_5^1 exp_i^2 + \beta_6^1 Northeast + \beta_7^1 Midwest + \beta_8^1 South + \varepsilon_i^1$$

9 Question 9

a To notice this first we would state that $H_0 : \beta_{fem} = \beta_{fem2} = 0$ and $H_a : \beta_{fem} \neq \beta_{fem2} \neq 0$. Where we get a p value of 0.4734 so it means that we fail to reject the null hypothesis.

b Now this is $H_0 : \beta_{femmarried} = 0$ and $H_a : \beta_{femmarried} \neq 0$. Where we get a p value of .0532 thus we fail to reject our null hypothesis, however, this is by very little since we are looking at the 5 percent significance level. We then could say that we fail to reject the idea that being married has the same effect on women and men.

c Now it is $H_0 : \beta_{schoolyr} = \beta_{exp}$ and $H_a : \beta_{schoolyr} \neq \beta_{exp}$. Where we get a p value of zero thus we reject the null hypothesis that they are the same. Thus we say that there is a difference between the return of a year of experience vs schooling.

d Now it is $H_0 : \beta_{married} + 25 * \beta_{marriedage} = 0$ and $H_a : \beta_{married} + 25 * \beta_{marriedage} \neq 0$. Where we get a p value of 0 thus again with strong evidence we reject the null hypothesis and can say that being married does have an impact on the wage of age 25 men.

e Now it is $H_0 : \beta_{college} = 10 * \beta_{exp}$ and $H_a : \beta_{college} \neq 10 * \beta_{exp}$. According to our p value of 0 we reject the null hypothesis thus we could say that the no 10 years of experiance does not have the same effect on wages as a college degree does.